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CSA0707

SIMATS ENGINEERING
ASSESSMENT TOOL 3

37

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared 1	Good understanding with minor gaps 1	Basic understanding; some concepts unclear 1	Poor understanding of concepts 1
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects 1	Adequate comparison with relevant points 1	Limited comparison; lacks depth 1	Minimal or incorrect comparison 1
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters 1	Mostly relevant criteria 1	Some irrelevant or missing criteria 1	Poor or no criteria selection 2
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification 1	Good analysis with some justification 1	Basic analysis with weak reasoning 1	No clear analysis 1
Organization & Structure	10%	Well-organized, clear, and logically structured 2	Generally organized 1	Somewhat disorganized 1	Poorly structured 1
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison 1	Uses some examples 2	Limited examples 1	No supporting examples 2
Clarity of Presentation	10%	Very clear, concise, and easy to understand 1	Mostly clear 2	Somewhat unclear 1	Difficult to understand 1
Conclusion & Insights	10%	Insightful conclusion with strong justification 1	Clear conclusion 1	Basic conclusion 1	No or weak conclusion 1
		9	10	8	10

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Star topology vs Bus topology — CO4–AL3

a. Star topology is more expensive because it needs a central switch / hub & more cabling.

b. Bus topology : Bus topology is cheaper because it uses a single backbone cable.

c. Reliability : Star topology is more reliable because failure of one nodes does not affect other nodes.

d. Bus failure : In bus topology, failures of the main backbone cable can stop the entire network.

e. performance : Star topology provides better performance because the central devices manage data transmission.

f. traffic : Bus topology may have more collisions & slower performance when more devices manages are added.

(22) Mesh topology vs star topology

- Mesh topology: It has very high fault devi[ce] because each node has multiple connections.
- Alternate paths: If one link fails, data c[an] travel through another available path.
- Reliability:- mesh topology provides very [high] reliability & continuous communication.

- star topology:- It has moderate fault tolerance because individual node failure do not affect other nodes.
- central Device:- If the hub (or) switch fa[ils] in star topology, the entire networks can st[op].
- conclusion:- mesh topology is more robust & suitable for critical networks than star topology.

topology vs Mesh topology - Installation.

is simple & easy to install.

- cable requirement : It uses a single backbone cable, reducing the number of connections.
- Bus topology is cheaper & suitable for small networks.
- mesh topology it is complex to install because nodes require multiple connections.

fault devices connecting.

4. Hybrid topology vs star topology - Scalability.

- Hybrid topology :- It combines two (or) more different topologies, providing high flexibility.
- Expansion :- New network segments can be added without greatly affecting existing segments.
- Scalability :- Hybrid topology can easily grow according to organizational requirements.
- star topology :- It is also scalable but depends on the capacity of the central switch (or) hub.

* port limit: when all ports are occupied, an additional switch (or) hardware may be required.

5) Maintenance of Different topologies.

* Star topology: It is easy to maintain because faults can be quickly identified & isolated.

* mesh topology: It is difficult to maintain because it has many connections & a complex structure.

* Bus topology:- It is harder to maintain because faults in the backbone cable can affect the whole network.

* trouble shooting:- finding faults in bus topology can take more time.

* Hybrid topology:- It has moderate maintenance complexity because it combines different topologies.